Secure Web Server on EC2

Project Overview

This project demonstrates deploying and securing a static website on AWS EC2 instances using Apache (httpd). It covers core cloud fundamentals: compute provisioning, network security, IAM best practices, and web server configuration — skills directly relevant to Cloud Support Engineer, Linux Administrator, and NOC Engineer roles.

Live Result: Two EC2 instances running the same secured static site, each verified independently.

Architecture

```
Internet
   |
   v
[Security Group: 22 (My IP only), 80 (Public)]
   |
   v
[EC2 Instances - Amazon Linux (x2)]
   |
   +-- Apache (httpd) Web Server
   |       |
   |       v
   |   /var/www/html/index.html
   |
   +-- IAM Role attached (no hardcoded credentials)
```

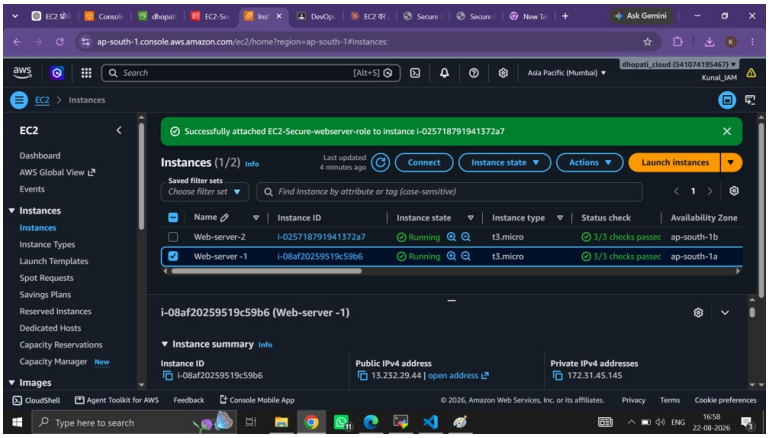
Tools and Services Used

Service	Purpose
Amazon EC2	Compute instances hosting the web server (2x, t3.micro)
Security Groups	Network-level firewall (inbound rules restricted to SSH, HTTP)
IAM Role	Attached to each instance for secure, credential-free AWS access
Apache (httpd)	Web server software serving the static page

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1. Launch EC2 Instances

- Amazon Linux AMI selected, instance type: **t3.micro** (Free Tier eligible)
- Two instances launched: **Web-server-1** and **Web-server-2**, both passing 3/3 status checks
- IAM Role (`EC2-Secure-webserver-role`) attached to both at launch



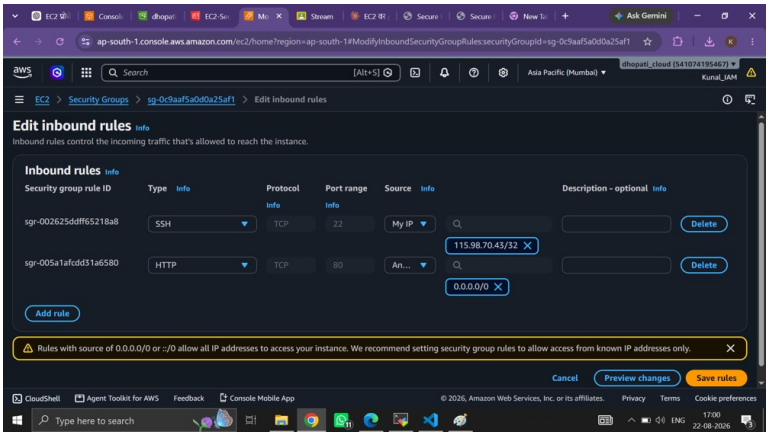
EC2 Instances Running

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2. Security Group Configuration

Type	Port	Source
SSH	22	My IP only (restricted)
HTTP	80	0.0.0.0/0 (public)

Restricting SSH to a specific IP (instead of `0.0.0.0/0`) significantly reduces brute-force and unauthorized access risk, while still allowing the website to be publicly reachable over HTTP.

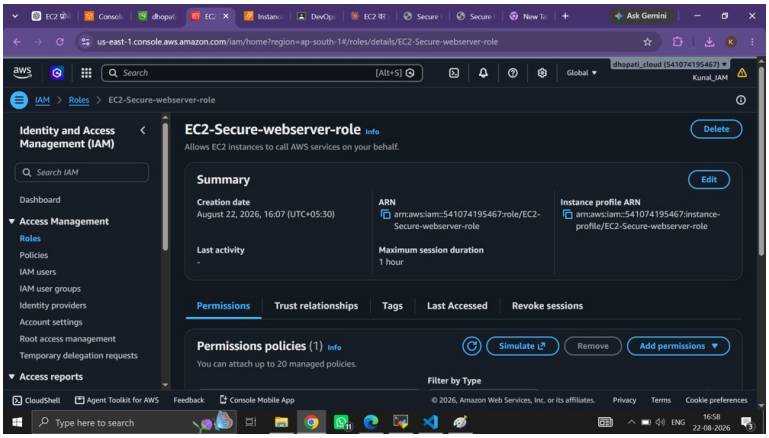


Security Group Inbound Rules

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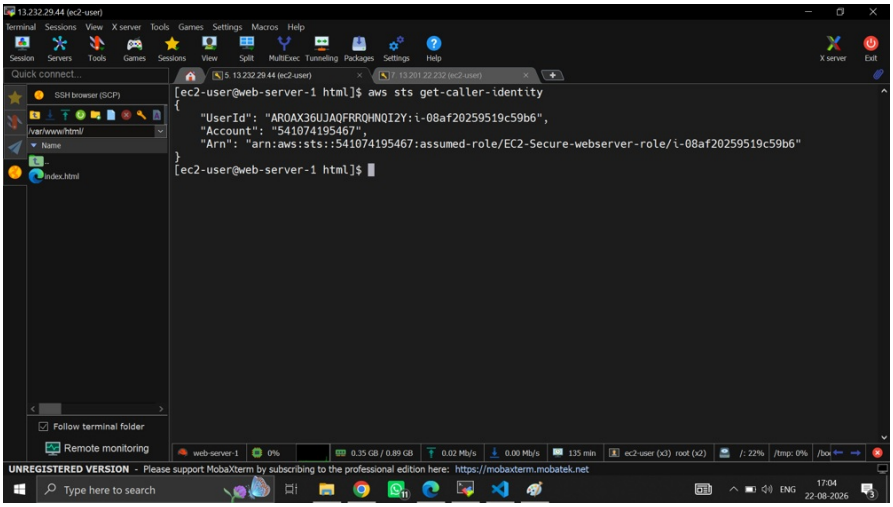
3. IAM Role (No Hardcoded Credentials)

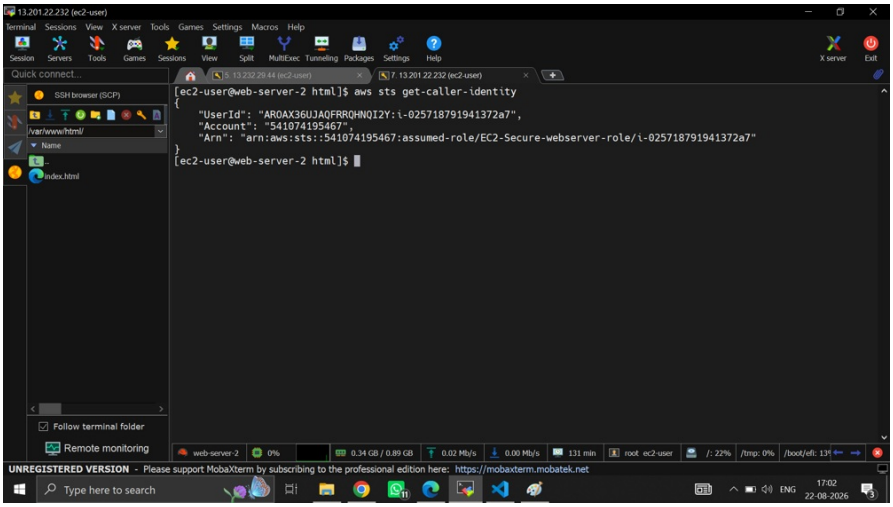
An IAM Role (`EC2-Secure-webserver-role`) was created and attached directly to both EC2 instances, allowing them to call AWS services securely without embedding access keys anywhere on the server.



IAM Role Summary

Verification — running `aws sts get-caller-identity` on each instance confirms the instance is authenticating via the assumed IAM role, not static credentials:





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4. Connect via SSH

```
ssh -i project-key.pem ec2-user@<13.232.29.44>
ssh -i project-key.pem ec2-user@<13.201.22.232>
```

5. Install Apache (httpd)

```
sudo yum update -y
sudo yum install httpd -y
```

6. Start and Enable the Service

```
sudo systemctl start httpd
sudo systemctl enable httpd
sudo systemctl status httpd
```

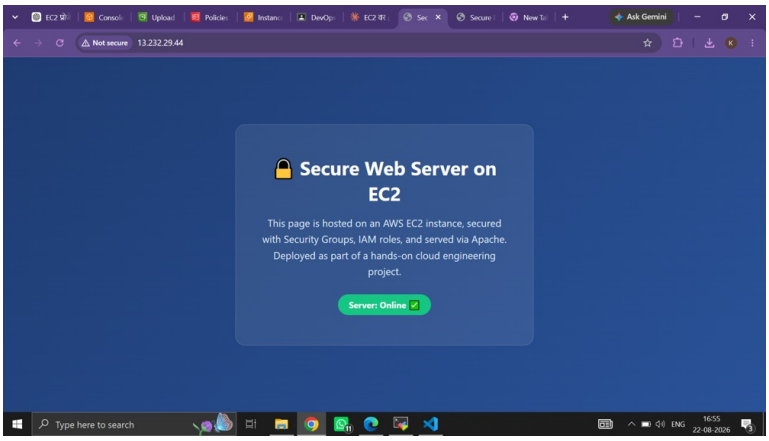
7. Deploy the Website

```
scp -i project-key.pem index.html ec2-
user@<13.232.29.44>:/var/www/html/
scp -i project-key.pem index.html ec2-
user@<13.201.22.232>:/var/www/html/

sudo mv /tmp/index.html /var/www/html/
sudo chown apache:apache /var/www/html/index.html
sudo chmod 644 /var/www/html/index.html
```

8. Verify

- Opened `http://<13.232.29.44>` in browser
- Opened `http://<13.201.22.232>` in browser
- Confirmed the page loads correctly with the web server active on both instances



Live Website Output

Source Code — `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">
<title>Secure EC2 Web Server</title>
<style>
  * { margin: 0; padding: 0; box-sizing: border-box; }
  body {
    font-family: 'Segoe UI', Arial, sans-serif;
    background: linear-gradient(135deg, #1e3c72, #2a5298);
    color: #fff;
    height: 100vh;
    display: flex;
    align-items: center;
    justify-content: center;
    text-align: center;
  }
  .card {
```

```
      background: rgba(255,255,255,0.08);
      border: 1px solid rgba(255,255,255,0.2);
      border-radius: 16px;
      padding: 50px 40px;
      max-width: 500px;
    }
    h1 { font-size: 2rem; margin-bottom: 15px; }
    p { font-size: 1.05rem; line-height: 1.6; color: #dce6f2; }
    .badge {
      display: inline-block;
      margin-top: 25px;
      padding: 8px 18px;
      background: #16c784;
      border-radius: 20px;
      font-size: 0.9rem;
      font-weight: bold;
    }
  </style>
  </head>
  <body>
    <div class="card">
      <h1>🔒 Secure Web Server on EC2</h1>
      <p>
        This page is hosted on an AWS EC2 instance, secured with
        Security Groups, IAM roles, and served via Apache.
        Deployed as part of a hands-on cloud engineering project.
      </p>
      <span class="badge">Server: Online ✅</span>
    </div>
  </body>
  </html>
```

Security Measures Applied

- Security Group restricted to only necessary ports (22, 80)
- SSH access limited to a specific IP rather than `0.0.0.0/0`
- IAM role used instead of hardcoded access keys
- File permissions set correctly on web root (`644`, owned by `apache`)

Outcome

Successfully deployed a live, secured static website across two EC2 instances, demonstrating end-to-end skills across compute, networking, IAM, and web server administration — with IAM role usage independently verified on each instance.

Possible Future Enhancements

- Add HTTPS using Let's Encrypt / Certbot or ACM + Application Load Balancer
- Map a custom domain using Route 53
- Automate provisioning using a Bash script or Terraform
- Add CloudWatch monitoring and alarms for the instances

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